

PRODUCT NAME	Shredder	EE TEAM
APPROVED BY		Fearghus Tyler
DATE OF APPROVAL	2026-06-01	Bao Nguyen
PART COUNT		62 Yaqoub Rabiah
TOTAL COST		\$969.88 Fox Kang

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
5KE182BC205B	AC Motor (3 HP)	GE 3-phase TEFC induction motor; 230 V (low-voltage), 1750 RPM, 7.6 A FLA, NEMA 182T frame. Bench-test motor.	1	EA	https://www.electricmotorsforless.com/	\$0.00	\$0.00
HY02D211B-T	VFD	Huanyang variable frequency drive, 110 V single-phase input, 220 V 3-phase output, 2.2 kW (3 HP). Selected via decision matrix.	1	EA	https://www.amazon.com/Huanyang-1-Phase-Variable-Frequency-Output-3hp/dp/B0DMF3GB38	\$200.00	\$200.00
APCS-300R30-AD	Brake Resistor	LS Electric 300 W, 30 ohm aluminum-clad braking resistor for VFD dynamic braking on P+/PR. Purchased from AutomationDirect. Note: an earlier identical-part-number unit (ATO branded) was ordered but never arrived after shipping; replaced with this LS Electric unit which is the brake resistor actually installed.	1	EA	https://www.automationdirect.com/adc/shopping/catalog/motion_control/ls_electric_l7c_l7ca_series/servo_systems/servo_accessories/apcs-300r30-ad	\$29.00	\$29.00
71008	Main Power Switch	Powertec 120 V ON/OFF main disconnect switch for entire control panel.	1	EA	https://powertecproducts.com/71008/	\$15.00	\$15.00
TMC 8 3C 15A (2907618)	VFD Branch Breaker	Phoenix Contact thermal-magnetic DIN-rail circuit breaker, 15 A, on VFD neutral leg (per as-built). Obtained at no cost from the PSU EPL free pile.	1	EA	https://www.phoenixcontact.com/	\$0.00	\$0.00
NDR-480-24	24 VDC Power Supply	Mean Well 480 W, 24 VDC DIN-rail power supply for PLC, HMI, indicator LEDs, and VFD control loops. Donated by sponsor / obtained at no cost to the EE team.	1	EA	https://www.meanwell.com/	\$0.00	\$0.00
SD-N35	Main Power Contactor	Mitsubishi 3-pole magnetic contactor. Energized through E-stop chain via E22B1 NC contact.	1	EA	https://www.ebay.com/itm/... (tpmachinery seller, order 23-14145-39817, Jan 2026)	\$12.59	\$12.59
EURO S4LH	Fuse Block (button common)	Morsettitalia fuse-holder terminal block. Holds 1 A fast-blow fuse protecting the shared pushbutton +24 V supply.	1	EA	https://www.morsettitalia.com/	\$8.00	\$8.00
EURO S10H-5H	Fuse Block (AC branches)	Morsettitalia fuse-holder terminal block for AC branch protection.	1	EA	https://www.morsettitalia.com/	\$12.00	\$12.00
H2-DM1E	PLC CPU / Ethernet	AutomationDirect Do-more CPU with built-in Ethernet, fits D2-09B-1 DL205 rack. Purchased Jan 2026 from planet-surplus as a USED combo for \$350 total, includes: D2-09B-1 chassis, D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2, and D2-FILL filler panels.	1	EA	https://www.ebay.com/itm/... (planet-surplus seller, order 20-14113-87209, Jan 2026)	\$350.00	\$350.00
D2-09B-1	PLC Base / Rack	AutomationDirect DirectLOGIC 205 9-slot AC-powered base. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
D2-08ND3	Discrete Input Module	AutomationDirect 8-ch, 24 VDC, sourcing/sinking discrete input module. Slot 2. Inputs X0-X7. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
F2-08AD-1	Analog Input Module	AutomationDirect 8-ch, 4-20 mA analog current input module. Slot 3. CH1+ reads VFD VO current monitor. Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
F2-04RTD	RTD Input Module	AutomationDirect 4-ch RTD input module. Slot 4. Installed but unused (spare). Included in the H2-DM1E used combo purchase.	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
D2-08TR	Relay Output Module	AutomationDirect 8-ch relay output module. Slot 5. Drives button LEDs (Y0-Y4) and VFD FOR/REV/RST (Y6/Y7/Y5). Included in the H2-DM1E used combo purchase (eBay listing showed D2-08TD2 but seller substituted D2-08TR which is what we use).	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00
F2-08DA-2	Analog Output Module	AutomationDirect 8-ch, 0-10 V analog voltage output module. Slot 6. +V1 drives VFD VI speed reference. Included in the H2-DM1E used combo purchase (eBay listing showed F2-02DA but actual unit is F2-08DA-2).	1	EA	https://www.automationdirect.com/	\$0.00	\$0.00

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
D2-FILL	Rack Filler Panels	AutomationDirect blank fill panels for unused rack slots 7-9. Included in the H2-DM1E used combo purchase.	3	EA	https://www.automationdirect.com/	\$0.00	\$0.00
EA1-T4CL	HMI Touch Panel	AutomationDirect C-More Micro-Graphic 4-inch color touch panel HMI. Purchased used from eBay for \$148.95 (Apr 2026). Purchase included the serial cable for HMI-to-CPU communication (RS-232 to H2-DM1E serial port).	1	EA	https://www.ebay.com/itm/306662618920	\$148.95	\$148.95
S2PR-P3G	Forward Pushbutton Head	Autonics 22 mm illuminated green pushbutton head (Forward command -> X0).	1	EA	https://www.autonics.com/us/model/S2PR-P3GAD	\$1.90	\$1.90
S2PR-P3Y	Reverse Pushbutton Head	Autonics 22 mm illuminated yellow pushbutton head (Reverse command -> X1).	1	EA	https://www.autonics.com/us/model/S2PR-P3YAD	\$1.90	\$1.90
S2PR-P3B	Run/Deadman Pushbutton Head	Autonics 22 mm illuminated blue pushbutton head (Run 1 -> X2, Run 2 -> X3).	2	EA	https://www.autonics.com/us/model/S2PR-P3BAD	\$1.90	\$3.80
S2PR-P3R	Reset Pushbutton Head	Autonics 22 mm illuminated red pushbutton head (planned Reset -> X6).	1	EA	https://www.autonics.com/us/model/S2PR-P3RAD	\$4.50	\$4.50
SA-CA	NO Contact Block (1NO)	Autonics finger-safe 1NO contact block. 110 VAC/10 A. Used with each S2PR pushbutton.	5	EA	https://www.autonics.com/us/model/SA-CA	\$2.80	\$14.00
SA-LDG	LED Block - Green (24 VDC)	Autonics modular LED indicator block, green, 12-24 VDC. Forward button illumination -> Y0.	1	EA	https://www.autonics.com/us/model/SA-LDG	\$4.50	\$4.50
SA-LDY	LED Block - Yellow (24 VDC)	Autonics modular LED indicator block, yellow, 12-24 VDC. Reverse button illumination -> Y1.	1	EA	https://www.autonics.com/us/model/SA-LDY	\$4.50	\$4.50
SA-LDB	LED Block - Blue (24 VDC)	Autonics modular LED indicator block, blue, 12-24 VDC. Run 1 and Run 2 illumination -> Y2, Y3.	2	EA	https://www.autonics.com/us/model/SA-LDB	\$4.50	\$9.00
SA-LDR	LED Block - Red (24 VDC)	Autonics modular LED indicator block, red, 12-24 VDC. Reset button illumination (planned).	1	EA	https://www.autonics.com/us/model/SA-LDR	\$4.50	\$4.50
E22B1	Emergency Stop Pushbutton	Cutler-Hammer / Eaton 22 mm mushroom E-stop, NC contact. In series with SD-N35 contactor coil.	1	EA	https://www.eaton.com/	\$25.00	\$25.00
Wamco 525	Fault Indicator Light	Wamco panel-mount LED indicator, 28 VDC, 0.6 W. Driven by PLC Y4 as machine-fault indicator.	1	EA	https://www.wamco.com/	\$15.00	\$15.00
Banana Socket (Blue)	Banana Socket Blue	Generic 4 mm panel-mount banana socket, blue. Quick-disconnect for VFD U output to motor T1. Obtained at no cost from a free pile.	1	EA		\$0.00	\$0.00
Banana Socket (White)	Banana Socket White	Generic 4 mm panel-mount banana socket, white. VFD V output to motor T2. Obtained at no cost from a free pile.	1	EA		\$0.00	\$0.00
Banana Socket (Red)	Banana Socket Red	Generic 4 mm panel-mount banana socket, red. VFD W output to motor T3. Obtained at no cost from a free pile.	1	EA		\$0.00	\$0.00
G1X1LG6	Slotted Wire Duct, 1"x1"	Panduit Type G wide-finger slotted PVC duct, light gray, 6 ft lengths.	6	FT	https://www.panduit.com/	\$3.43	\$20.58
C1LG6	Duct Cover, 1"W	Panduit duct cover for G1X1LG6, light gray, 6 ft lengths.	6	FT	https://www.panduit.com/	\$1.20	\$7.20
18 AWG Blue	Control Wire, Blue	18 AWG stranded blue control wire (NFPA 79 §13.2 DC control). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
18 AWG White	Control Return Wire	18 AWG stranded white wire for DC return (NFPA 79 §13.2). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
12 AWG Black	Power Wire, Black	12 AWG black wire for AC mains and motor leads. Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
12 AWG Green/Yellow	Ground Wire	12 AWG green/yellow stranded equipment grounding wire (NFPA 79). Obtained at no cost (some from team member's existing stock, some from free pile).	1	Roll		\$0.00	\$0.00
GMA-500mA or 0217.500MXP	Fuse - 500 mA F	Bussmann or Littelfuse 5x20 mm fast-blow fuse, 500 mA, 250 V. (Currently not installed per as-built note - retained as recommended addition.)	1	EA		\$1.50	\$1.50
GMA-1A or equiv.	Fuse - 1 A F	5x20 mm fast-blow fuse, 1 A, 250 V. Protects shared +24 V pushbutton common (in EURO S4LH).	1	EA		\$1.50	\$1.50
LM2596	DC-DC Buck Converter (24V to 5V/3.3V)	Seloky LM2596-based adjustable DC-DC buck converter module with voltmeter display (sold as 2-pack). Steps down 24 VDC panel supply to the input voltage required by the capacitive touch module.	1	Pack	https://www.amazon.com/Converter-Adjustable-Regulator-Voltmeter-Display/dp/B0F1M2SR31	\$7.99	\$7.99

PART NUMBER	PART NAME	DESCRIPTION	QUANTITY	UNITS	LINK	UNIT COST	AMOUNT
SEB001	Control Panel Enclosure	Vorksmen industrial control enclosure (steel) housing the entire electrical system including PLC rack, VFD, contactor, PSU, terminal blocks, and operator panel.	1	EA	https://www.amazon.com/	\$59.98	\$59.98
AE14921a	3V Relay	AEDIKO 3 VDC PCB-mount relay. Used in the capacitive touch module sub-circuit to interface the low-voltage touch sensor output with the 24 VDC panel control voltage.	1	EA	https://www.amazon.com/	\$6.99	\$6.99
DGN-001	Bao's Dignity	Lost on the procurement loop after lowballing roughly every vendor between	1	EA		0	0
TOTAL PARTS			62			TOTAL	\$969.88

PART NUMBER	PART NAME	DESCRIPTION	FROM BOM FILE	REASON NOT USED
PE145TC-2-4	Motor (alternate)	Original BOM candidate motor	BOM_02_20_26	Not used; GE 5KE182BC205B used instead for bench testing
NBE7	Circuit Breaker	Original BOM candidate breaker	BOM_1_29 / BOM_02_20_26	Never purchased; replaced by Phoenix Contact TMC 8 3C 15A
(30 A fuse, TBD)	30 A Branch Fuse	Per old BOM, 2x 30 A fuses budgeted	BOM_1_29	Never needed in final design
1866-1441-ND	DC Power Supply	120/24 VDC PSU from Digi-Key	BOM_1_29	Superseded by Mean Well NDR-480-24
(generic)	VEVOR 3HP 120V VFD	Hobby-grade VFD option	VFD_120V_3HP_Decision_Matrix	Rejected during selection (hobby-grade, limited documentation)
(generic)	ATO 3HP 120V VFD	Higher-quality VFD option	VFD_120V_3HP_Decision_Matrix	Too expensive (\$450-500) for project budget
(generic)	110V 20A 3HP VFD	Mid-tier VFD option	VFD_120V_3HP_Decision_Matrix	Huanyang HY02D211B-T selected instead
(WEG single-phase 3HP)	WEG Single Phase Motor	1-phase general purpose 3 HP motor	Motor Spec	Considered but GE 5KE182BC205B used
NT3-18-213T	Worldwide Electric Motor	3-phase TEFC 3 HP motor	Motor Spec	Considered but not purchased
(VEVOR 5HP 230V)	VEVOR High Torque Motor	5 HP 1-phase compressor-style motor	Motor Spec	Considered but not purchased
L510 (2HP / 3HP)	TECO L510 VFD	VFD comparison option	Motor Spec	Considered but Huanyang selected
G0532	Marathon Gearmotor	Integrated gearbox motor	Motor Spec	Considered but not purchased
KBWC-15K	KB Electronics speed control	Triac speed control fan-style	Motor Spec	VFD approach used instead
TTP223B	Capacitive Touch Sensor	Single-channel cap touch	BOM_1_30_beta (safety draft)	Does not meet industrial standards
CAP1293-1-SN	Capacitive Touch Chip	3-channel cap touch chip	BOM_1_30_beta (safety draft)	Chip only; needs interface board
1864-1032-ND	Capacitive Proximity Sensor	Cap proximity sensor	BOM_1_30_beta (safety draft)	Too expensive (\$109.25)
F3S-TGR-NMPC-21-10	Omron Magnetic Safety Sensor	Non-contact magnetic safety switch	BOM_1_30_beta (safety draft)	Too expensive; out of stock
RSS 36-I2-D-R-ST5	Schmersal RFID Safety Sensor	RFID safety sensor	BOM_1_30_beta (safety draft)	Suspected industrial-only / non-retail
F3SG-SR / PG	Omron Safety Light Curtain	Safety light curtain	BOM_1_30_beta (safety draft)	Out of scope for capstone budget
(Ichor-provided)	Red/Green Indicator Lights	Sponsor-provided lights	BOM_02_20_26	Not used; Wamco 525 fault light and button LEDs cover indication
(Ichor-provided)	E-stop Pushbutton	Sponsor-provided E-stop	BOM_02_20_26	Replaced with Cutler-Hammer/Eaton E22B1 procured by team
V30AE000055	Insulated Ferrules, 12 AWG (gray)	Z+F crimp ferrules, 12 AWG, 10 mm length, gray (DIN color code). 500-piece pack.	URS Electronics invoice (purchased but not used)	Purchased from URS Electronics but ultimately not used in the as-built panel. Spare stock retained for future projects.